

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

SP8 - General Insurance : Pricing

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) With reference to components and mechanics of financial statements of a general insurance specify the following:

- i) List down the types of technical reserves. [2]
- ii) Describe the main sources of the funds and their various uses as reflected in the balance sheet of a general insurance company in detail.

[6]
[8 Marks]

Q.2) There is a rapid adoption of Artificial Intelligence (AI) in the insurance industry. Discuss the implications of the same in the context of :

- i) What are the key risks faced by organisations from the adoption of Artificial Intelligence, both from a people and an operational standpoint? [4]
- ii) Discuss how demand for various general insurance products is expected to change and the underlying reasons for such shifts, with reference to the growing use of Artificial Intelligence. [4]

[8 Marks]

Q.3) You are consulting a large Global Financial Institution to set up a General Insurance Company in India.

- i) List down the key considerations related to regulatory and economic factors and on which the Company needs to be advised? [5]
- ii) The Company wants to use Reinsurance as an effective tool of risk mitigation. What are the various types of reinsurance contracts that should be in consideration? [2]
- iii) What could be possible reasons that the regulator may not allow capital gearing reinsurance arrangements? [3]

iv)

- a) Explain what are Run-off reinsurance arrangements?
- b) Which products and lines of business are more suitable for run-off reinsurance covers?
- c) If a direct insurance has written a Rs 100 Cr portfolio from 1st April 2025 upto 31st March 2026, and borne a Rs 90 Cr loss until 30th June 2026. The insurer expects incremental losses to be between Rs 2 cr and Rs 80 Cr with 85% confidence

Should the insurer consider transferring this portfolio to a reinsurer. Why?

[6]

[Total : 16 Marks]

Q.4) You have joined as a pricing actuary in a new General Insurance Company being established in India. The company is in the process of setting up a new data system and the IT head has asked for your inputs on Data Governance

- i) What are the business and regulatory requirements that need to be addressed using data available in an insurance company? [2]
- ii) What are the flexibilities or functionalities required that the new data system needs to cater for a General Insurance company? [6]

[Total : 8 Marks]

Q.5) What are the major risks that an insurance company is exposed to in the following insurance products:

- Motor Third Party business
- Retail Health Indemnity Products
- Drone Insurance
- Specialized insurance like Footballer's leg or Singer's Larynx
- State Crop Scheme

[5 Marks]

Q.6) What are the key risks and uncertainties that an Indian General Insurance company would face during Global Political uncertainties? **[7 Marks]**

Q.7) You have been assigned a task of pricing a new product in a medium sized General insurance company

- i) What are the factors that determine which pricing methodology should be used? [2]
- ii) You are pricing for a large Marine Cargo insurance client. Below is the data available for its claims experience from the broker. Calculate if the client has been profitable and the final premium to be paid by the customer for it in the forthcoming year considering the exposure at 2000 units.

Policy Year	Premium	Claims	Exposure
2022-23	100	100	200
2023-24	200	160	500
2024-25	300	300	800
2025-26	400	450	1500

Assume annual inflation of 10% p.a., brokerage of 7.5%, 0% profit margin of premium and no other expenses applicable.

[6]

- iii) There is an opportunity to cede out the risk partially to a reinsurer against 5% RI commission. Should you cede out 50% of the risk? **[4]**

[Total : 12 Marks]

Q.8) A newly joined actuarial student is curious about the pricing methodology followed for Property business

- i) He is confused between Exposure and Experience based pricing. Specify when each of the above is used? [2]
- ii) The student has observed the pricing exercise and noticed that the prices at renewals have been reducing in spite of claims experience been similar as compared to previous years. Explain the possible reasons for the same. [6]
- iii) The company is using internally developed exposure curves for exposure rating. What are key considerations while using these exposure curves for future new business? [4]

[Total: 12 Marks]

Q.9) Explain the concept of a catastrophe model, describe its structure, and list the key perils that can be modelled using it. [6 Marks]

Q.10) You are a Reinsurance consultant to a large General Insurance company, focusing on Property Insurance. The Company's CEO has asked to you optimize its current Reinsurance structure even at the cost of complicating it.

The Company categories its policies into various cohorts basis size, hazard and price sensitivity. According to the Company's risk appetite, small risk is below 50 Cr and large risk is anything above 100 Cr PML. Hazard categories used are Low, Medium and High and underwriting cycle could soft or hard. Describe the ideal reinsurance arrangement for each cohort, while covering the below terminologies:

He wants to understand the following terms:

- Inuring reinsurance
- Re-instatement
- Deductible and aggregate deductible
- Indexation clauses
- Swing-rated contracts
- Loss ratio caps

[10 Marks]

Q.11) The Company wants you to review the current rates used for Motor Insurance.

- i) Compare the common forms of Supervised Machine Learning based on when each one is used. [3]
- ii) The Company divides its portfolio into 4 risk cohorts. Calculate the Burn Cost to be charged for each of the cohorts using the following data:

Base Claims Frequency : 25%

Base Claims severity : 35,000

Count relativities table:

Car Category	Age: Upto 5 Yrs	Age : 6-10 Yrs
Sedan	1	1.2
SUV	1.05	1.3

Amount relativities table:

Car Category	Age: Upto 5 Yrs	Age : 6-10 Yrs
Sedan	0.9	1.2
SUV	1	1.5

[5]

[Total : 8 Marks]
